

Partial Solution Packet

Nondense-square Banach algebras have a representation with non-complemented essential space

Source: arXiv:1703.00882, Question 3.4

Source Question

Gardella–Thiel ask whether every Banach algebra A without a left unit has a representation $\varphi : A \rightarrow B(X)$ on some Banach space X whose essential space is not complemented.

The essential space of φ is

$$X_\varphi = \overline{\text{span}}\{\varphi(a)x : a \in A, x \in X\}.$$

Result

Theorem. *Let A be a Banach algebra such that $\overline{A^2} \neq A$. Then there is a Banach space X and a bounded representation $\varphi : A \rightarrow B(X)$ such that X_φ is not complemented in X .*

Proof. Since $\overline{A^2} \neq A$, Hahn–Banach gives a nonzero functional $\chi \in A^*$ such that $\chi(ab) = 0$ for all $a, b \in A$.

Let

$$0 \longrightarrow H \xrightarrow{i} Z_2 \xrightarrow{q} H \longrightarrow 0$$

be the Kalton–Peck nonsplit twisted Hilbert exact sequence [1]. Thus H is a Hilbert space, $i(H)$ is a closed subspace of Z_2 , $qi = 0$, and $i(H)$ is not complemented in Z_2 . Put $X = Z_2$ and

$$T = iq \in B(X).$$

Then $T^2 = iqiq = 0$, and $\text{Range}(T) = i(H)$. The range is not complemented: if it were, the quotient map $q : Z_2 \rightarrow H$ would admit a bounded right inverse by restricting q to a complement of $i(H)$, contradicting nonsplitting.

Define

$$\varphi(a) = \chi(a)T \quad (a \in A).$$

This is bounded and linear. Moreover, for $a, b \in A$,

$$\varphi(a)\varphi(b) = \chi(a)\chi(b)T^2 = 0$$

while

$$\varphi(ab) = \chi(ab)T = 0.$$

Hence φ is a representation.

Choose $a_0 \in A$ with $\chi(a_0) \neq 0$. Then

$$X_\varphi = \overline{\text{span}}\{\chi(a)Tx : a \in A, x \in X\} = \text{Range}(T) = i(H),$$

which is not complemented in X . This proves the claim. $\square$

Scope

The hypothesis $\overline{A^2} \neq A$ automatically excludes left units, since a left unit would imply $A = A^2$. The theorem therefore gives a positive answer to Question 3.4 for all Banach algebras with nondense square, including all nonzero Banach algebras with zero product.

The dense-square case remains outside this packet.

References

- [1] N. J. Kalton and N. T. Peck, *Twisted sums of sequence spaces and the three space problem*, Trans. Amer. Math. Soc. 255 (1979), 1–30.